in $(D_2, \mathfrak{D}_2)$*, where* Y *is a zero mean Gaussian process given by*

$$Y(s,t) = t\mathcal{W}_\Gamma(s) + s\mathcal{W}_\Gamma(t), \qquad 0 \le s,t \le 1,$$

with $\mathcal{W}_\Gamma = \sqrt{\Gamma}B$ *for a standard Brownian motion* B *and* $\Gamma \coloneqq \sum_k \operatorname{Cov}(h_1(X_0), h_1(X_k))$.

Proof of Theorem 2.3. We start with the decomposition given in (2.4)

$$\begin{aligned}\sqrt{n}U_n(h)(t,s) &= \sqrt{n}a_n(s,t)\theta + \sqrt{n}U_n(h_2)(s,t)\\ &\quad + \frac{1}{\sqrt{n}}\sum_{i=1}^{\lfloor ns\rfloor} h_1(X_i)\ b_{n,i}(t) + \frac{1}{\sqrt{n}}\sum_{j=1}^{\lfloor nt\rfloor} h_1(X_j)\ b_{n,j}(s).\end{aligned}$$

Clearly, $\sup_{t\ge 2/n}\max_{i\le n}|b_{n,i}(t)-t| \le 3/(n-1) \to 0$ as $n\to\infty$. Also, note that $(h_1(X_t))_t$ is again an L^2-m-approximable one-dimensional process: We have due to the ϱ-Hölder continuity of h_1 and the condition in (2.7)

$$\begin{aligned}\sum_{m=1}^{\infty}\mathbb{E}[(h_1(X_0)-h_1(X_0^{(m)}))^2]^{1/2} &\le \sum_{m=1}^{\infty} L\ \mathbb{E}[d(X_0,X_0^{(m)})^{2\varrho}]^{1/2}\\ &\le L\sum_{m=1}^{\infty} m\ \mathbb{E}[d(X_0,X_0^{(m)})^{p\varrho}]^{1/p} < \infty.\end{aligned}$$

Moreover, $\mathbb{E}[h_1(X_t)] = 0$ and we have that $\mathbb{E}[d(X_t,x)^2] < \infty$ for some $x \in M$ by Assumption 2.1, thus,

$$\begin{aligned}\mathbb{E}[|h_1(X_t)|^2] &\le 2\mathbb{E}[|h_1(X_t)-h_1(x)|^2] + 2h_1(x)^2\\ &\le 2L^2\mathbb{E}[d(X_t,x)^{2\rho}] + 2h_1(x)^2 < \infty.\end{aligned}$$

Hence, an application of Theorem 2.4 yields that we have the following functional weak convergence in the Skorokhod space $(D_2, \mathfrak{D}_2)$

$$\left(\frac{1}{\sqrt{n}}\sum_{i=1}^{\lfloor ns\rfloor} h_1(X_i)b_{n,i}(t) + \frac{1}{\sqrt{n}}\sum_{j=1}^{\lfloor nt\rfloor} h_1(X_j)\ b_{n,j}(s)\middle|\ 0\le s,t\le 1\right) \Rightarrow Y.$$

Furthermore observe that $\sup_{s,t\ge 2/n}\sqrt{n}|a_n(s,t)-st|\cdot|\theta| \le 3|\theta|\sqrt{n}/(n-1) \to 0$ as $n\to\infty$. Indeed, if $s\le t$, then $\sqrt{n}(a_n(s,t)-st)$ is bounded from above by

$$\sqrt{n}\left(\frac{ns(nt-1)}{n(n-1)} - st\right) = \sqrt{n}\frac{s(t-1)}{n-1} \to 0$$

also if $s \le t$, then $\sqrt{n}(a_n(s,t)-st)$ bounded from below by

$$\sqrt{n}\left(\frac{(ns-1)(nt-2)}{n(n-1)} - st\right) = \sqrt{n}\frac{s(t-1)}{n-1} + \sqrt{n}\frac{2/n-s-t}{n-1} \to 0.$$

If $t < s$, the calculations work in a similar fashion.

Thus it remains to show that the degenerate part $Z_n := \sqrt{n}U_n(h_2)$ converges weakly in $(D_2, \mathfrak{D}_2)$ to the zero process. To that end we make use of the convergence criteria laid out in detail in Bickel and Wichura [4]: We use the corollary given after Theorem 2 combined with the tightness condition of Theorem 3 in the very paper. More precisely, we verify that

(1) the finite-dimensional distributions of Z_n converge to those of the zero process;
(2) $\mathbb{E}[|Z_n(A)||Z_n(B)|] \leq C|A|^{3/4}|B|^{3/4}$ for all *neighboring* blocks $A, B \subseteq [0,1]^2$ for a certain constant $C \in \mathbb{R}_+$.

Here, two blocks $A = (s_1, t_1]\times(s_2, t_2]$, $0 \leq s_i < t_i \leq 1$, and $B = (u_1, v_1]\times(u_2, v_2]$, $0 \leq u_i < v_i \leq 1$, are said to be neighboring if they have a common edge. Adapting the notation in Bickel and Wichura [4] we write

$$X(A) := X(s_1, s_2) - X(t_1, s_2) - X(s_1, t_2) + X(t_1, t_2) \tag{2.11}$$

for the 2-dimensional increment of a process $X = (X(s,t))_{s,t\in[0,1]} \in D_2$ over a block A.

Statement (1): Observe that for $m \in \mathbb{N}$, $\alpha_1, \dots, \alpha_m \in \mathbb{R}$ and $\varepsilon > 0$ we have as a consequence of Corollary 2.6 and Markov's inequality

$$\begin{aligned}
&\mathbb{P}\Big(\Big|\sum_{k=1}^{m} \alpha_k Z_n(s_k, t_k)\Big| \geq \varepsilon\Big) \leq \varepsilon^{-1}\ \mathbb{E}\Big[\Big|\sqrt{n}\sum_{k=1}^{m} \alpha_k U_n(h_2)(s_k, t_k)\Big|\Big] \\
&\leq \varepsilon^{-1}\ \mathbb{E}\Big[\Big|\sqrt{n}\sum_{k=1}^{m} \alpha_k U_n(h_2)(s_k, t_k)\Big|^2\Big]^{1/2} \leq \frac{\sqrt{n}}{\varepsilon} \quad \sum_{k=1}^{m} |\alpha_k|\ \mathbb{E}[|U_n(h_2)(s_k, t_k)|^2]^{1/2} \\
&= \frac{\sqrt{n}}{\varepsilon}\ \sum_{k=1}^{m} |\alpha_k|\ \mathbb{E}\Big[\Big(\frac{1}{n(n-1)}\sum_{i=1}^{\lfloor ns_k\rfloor}\sum_{\substack{j=1\\ i\neq j}}^{\lfloor nt_k\rfloor} h_2(X_i, X_j)\Big)^2\Big]^{1/2} \\
&\leq \frac{\sqrt{n}}{\varepsilon}\ \sum_{k=1}^{m} \frac{|\alpha_k|}{n(n-1)} \left(\sum_{\substack{i,j=1\\ i\neq j}}^{n}\ \sum_{\substack{u,v=1\\ u\neq v}}^{n} \big|\mathbb{E}[h_2(X_i, X_j)h_2(X_u, X_v)]\big|\right)^{1/2} \lesssim \frac{\|\alpha\|_1}{\varepsilon\sqrt{n}}.
\end{aligned}$$

Hence it follows from the Cramér-Wold device that the finite dimensional distributions of Z_n converge weakly to those of the zero process.

Statement (2): Relying on the Cauchy-Schwarz inequality, it is sufficient to verify the moment condition

$$\mathbb{E}[|Z_n(A)|^2] \leq C|A|^{3/2} \tag{2.12}$$

for all blocks A in the unit square. This condition is verified in Theorem 2.5. $\square$

The proofs of the next two theorems are postponed to Section 3.3.

Theorem 2.4 (Two-parameter Donsker type FCLT). *Let $Z = (Z_t)_{t\in\mathbb{Z}}$ be a real-valued L^2-m-approximable process such that $\mathbb{E}[Z_t] = 0$ and $\mathbb{E}[Z_t^2] < \infty$. Define the partial-sum process*

$$Y_n \colon [0,1]^2 \to \mathbb{R}, \quad (s,t) \mapsto \frac{t}{\sqrt{n}} \sum_{i=1}^{\lfloor ns \rfloor} Z_i + \frac{s}{\sqrt{n}} \sum_{i=1}^{\lfloor nt \rfloor} Z_i. \tag{2.13}$$

Then $Y_n \Rightarrow Y$ in the two-dimensional Skorokhod space $(D_2, \mathfrak{D}_2)$, where the Gaussian process Y is given by

$$Y(s,t) = t\mathcal{W}_\Gamma(s) + s\mathcal{W}_\Gamma(t)$$

with $\mathcal{W}_\Gamma = \sqrt{\Gamma}B$, for a standard Brownian motion B and $\Gamma = \sum_{h\in\mathbb{Z}} \mathrm{Cov}(Z_0, Z_h)$.

Theorem 2.5 (Moment condition for $U_n(h_2)$). *Granted Assumption 2.1 (p,q,ρ) is satisfied there is a universal constant $C \in \mathbb{R}_+$ such that for all blocks $A = (s,u] \times (t,v] \subset [0,1]^2$*

$$\mathbb{E}\left[\left| \sqrt{n}\, U_n(h_2)(A) \right|^2 \right] \le C|A|^{3/2}, \tag{2.14}$$

where $U_n(h_2)(A)$ is the 2-dimensional increment of $U_n(h_2)$ over A in the sense of (2.11), i.e.,

$$U_n(h_2)(A) = U_n(h_2)(u,v) + U_n(h_2)(s,t) - U_n(h_2)(u,t) - U_n(h_2)(s,v).$$

The proof of Theorem 2.5 yields a straightforward corollary:

Corollary 2.6 (Second moment of $U_n(h_2)$). *Granted Assumption 2.1 (p,q,ρ) is satisfied there is a universal constant $C \in \mathbb{R}_+$, such that*

$$\sum_{\substack{1\le i<j\le n,\\ 1\le k<\ell\le n}} |\mathbb{E}[h_2(X_i,X_j)h_2(X_k,X_\ell)]| \le Cn^2.$$

3. Mathematical details

3.1. Two sample test for Fréchet variances

Proof of Theorem 1.2. The proof is divided into two steps.

Step 1. We verify the assumptions of Theorem 3.1 for $X_i := W_r^2(\mathrm{PD}(\mathcal{X}_i), \mu_\mathcal{X})$, resp., $Y_i := W_r^2(\mathrm{PD}(\mathcal{Y}_i), \mu_\mathcal{Y})$. We show that $(X_i)_i$ and $(Y_i)_i$ inherit the L^p-m-approximation property from the underlying point cloud processes $(\mathcal{X}_t)_t$ and $(\mathcal{Y}_t)_t$, which then implies a strong law of large numbers as well as an invariance principle for the corresponding partial sum processes. Again, we only present the calculations for $(X_i)_i$: To that end, let

$$X_i^{(m)} := W_r^2(\mathrm{PD}(\mathcal{X}_i^{(m)}),\ \mu_\mathcal{X}), \qquad m \in \mathbb{N}$$